Chapter 1 Introduction

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Data Layout

- In Multivariate Data, there will be n experimental or sampling units, each having p variables/characteristics being observed/measured
- The data will be structured in an array of n rows (units) and p columns (variables), which is labelled as $\mathbf{X}$
- x_{jk} represents the measurement of the kth variable on the jth unit

$$\mathbf{X} = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1,p-1} & x_{1p} \\ x_{21} & x_{22} & \cdots & x_{2,p-1} & x_{2p} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ x_{n-1,1} & x_{n-1,2} & \cdots & x_{n-1,p-1} & x_{n-1,p} \\ x_{n1} & x_{n2} & \cdots & x_{n,p-1} & x_{np} \end{bmatrix}$$

Example

Example: Height, weight and age are measured for 23 people. Then p = 3 and n = 23.

$$\mathbf{X} = \begin{bmatrix} x_{1,1} & x_{1,2} & x_{1,3} \\ x_{2,1} & x_{2,2} & x_{2,3} \\ \vdots & \vdots & \vdots \\ x_{22,1} & x_{22,2} & x_{22,3} \\ x_{23,1} & x_{23,2} & x_{23,3} \end{bmatrix}$$

Mean and variance

- Means:

$$x_k = \frac{1}{n}\sum_{j=1}^{n} x_{jk} \quad k = 1, ..., p$$

- Sums of Squares:

$$w_{ik} = \sum_{j=1}^{n} \left(x_{ji} - \overline{x}_i\right)\left(x_{jk} - \overline{x}_k\right) \quad i = 1, ..., p;\ k = 1, ..., p$$

- Variances:

$$s_k^2 = s_{kk} = \frac{1}{n}\sum_{j=1}^{n} \left(x_{jk} - \overline{x}_k\right)^2 = \frac{w_{kk}}{n} \quad k = 1, ..., p$$

(Note dividing by n, not $n-1$ for now)

Covariance and correlation

- Correlations:

$$r_{ik} = \frac{s_{ik}}{\sqrt{s_{ii}s_{kk}}} = \frac{\sum_{j=1}^{n}(x_{ji}-\overline{x}_i)(x_{jk}-\overline{x}_k)}{\sqrt{\sum_{j=1}^{n}(x_{ji}-\overline{x}_i)^2\sum_{j=1}^{n}(x_{jk}-\overline{x}_k)^2}} = \frac{w_{ik}}{\sqrt{w_{ii}w_{kk}}}$$

$$i = 1, \ldots, p;\ \ k = 1, \ldots, p$$

- Covariances:

$$s_{ik} = \frac{1}{n}\sum_{j=1}^{n}(x_{ji}-\overline{x}_i)(x_{jk}-\overline{x}_k) = \frac{w_{ik}}{n} \qquad i = 1, \ldots, p;\ \ k = 1, \ldots, p$$

(Note dividing by n, not $n-1$ for now)

Descriptive statistics: arrays

- Mean (Column) Vector:

$$\underset{p\times 1}{\overline{\mathbf{x}}} = \begin{bmatrix} \overline{x}_1 \\ \vdots \\ \overline{x}_p \end{bmatrix}$$

- Variance - Covariance Matrix:

$$\underset{p\times p}{\mathbf{S}_n} = \begin{bmatrix} s_{11} & \cdots & s_{1p} \\ \vdots & \ddots & \vdots \\ s_{p1} & \cdots & s_{pp} \end{bmatrix} \quad \text{with } s_{ik} = s_{ki}$$

- Correlation Matrix:

$$\underset{p\times p}{\mathbf{R}} = \begin{bmatrix} 1 & \cdots & r_{1p} \\ \vdots & \ddots & \vdots \\ r_{p1} & \cdots & 1 \end{bmatrix} \quad \text{with } r_{ik} = r_{ki}$$

Graphical methods

"A picture is worth a thousand words..."

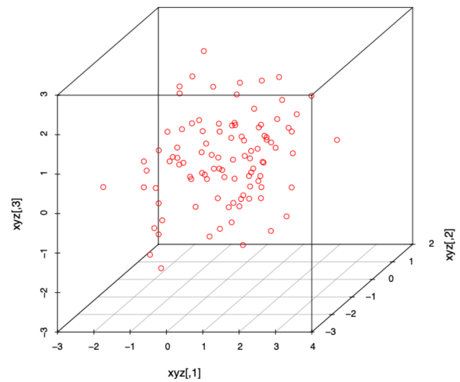
Some useful approaches for visualizing multivariate data are:

- 3D plots
- Scatter plots
- Bubble plots
- Stars
- Chernoff faces

(on the other hand, 1001 words are worth more than a picture...)

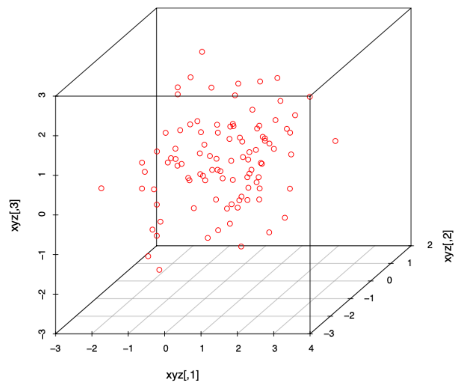
3D plots

Three dimensional scatter plots:



3D plots

Three dimensional scatter plots:

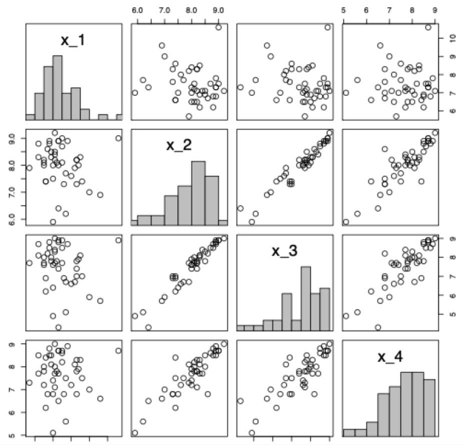


3D plots

- Often a good choice for three dimensional data.
- Enables us to see patterns that would disappear if the data was projected to two dimensions.
- Best when interactive (i.e. when the plot can be rotated).

Scatter plots

Scatter plots of each pair of variables, usually combined with the marginal histograms.

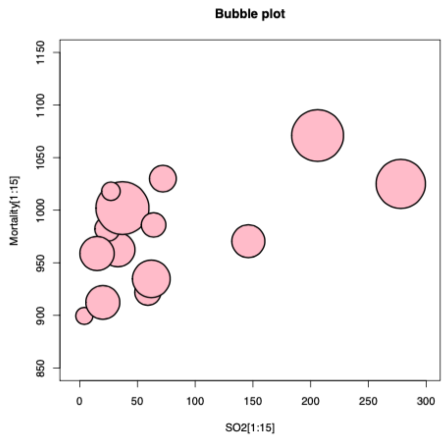


Scatter plots

- Can, unlike 3D plots, be used for p greater than 3.
- Usually a good first choice for plotting.
- Good for detecting dependencies between pairs of variables.
- Higher-dimensional perspective is lost – important dependencies between more than two variables might go unnoticed.
- Useful for assessing multivariate normality.

Bubble plots

In a regular 2D plot, a third dimension is illustrated using bubbles of different sizes.

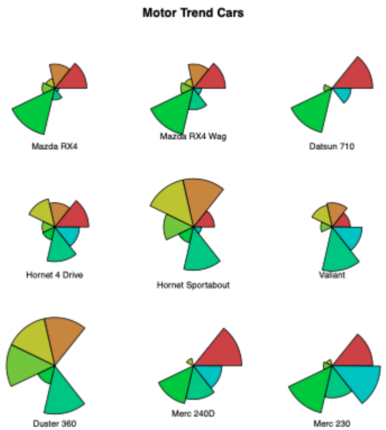


Bubble plots

- Simple, but easy to interpret.
- Possible to make nice-looking plots.
- Possible extensions to higher dimensions?
  - 3D bubble plots, colours of bubbles, shapes of bubbles, time dimension...

Stars

The lengths of the rays from the center of the figure represent the values of the variables. Example with $p = 7$ and $n = 9$:



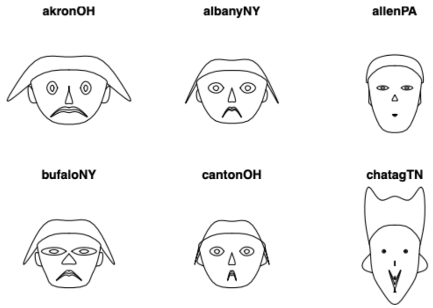
Stars

- Can be used for dimensions higher than three.
- Relatively easy to interpret.
- Can be useful for finding similar data points.
- If plotted in a time sequence, can illustrate change over time.
- Only useful for relative comparisons.

Chernoff faces

The human mind is extremely good at facial recognition. Chernoff proposed illustrating data sets with facial features.

Herman Chernoff (1973), The use of faces to represent points in k-dimensional space graphically, Journal of the American Statistical Association, 68, pp. 361-368.



Chernoff faces

- Each variable is represented by a facial feature (e.g. length of nose, size of eyes, width of head...).
- Easy to see groups or clusters in the data.
- Can be used to find outliers.
- If plotted in a time sequence, can illustrate change over time.
- Can be used for $p \leq 18$.
- Care must be taken when choosing which variable is represented by which facial feature. We react stronger to changes in some of the features!

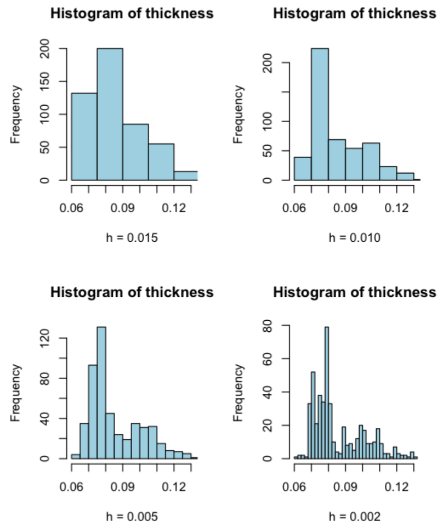
Example - 1D Hidalgo Stamp Data

Example - 1D Hidalgo Stamp Data. We use data "stamp" in R package BSDA.

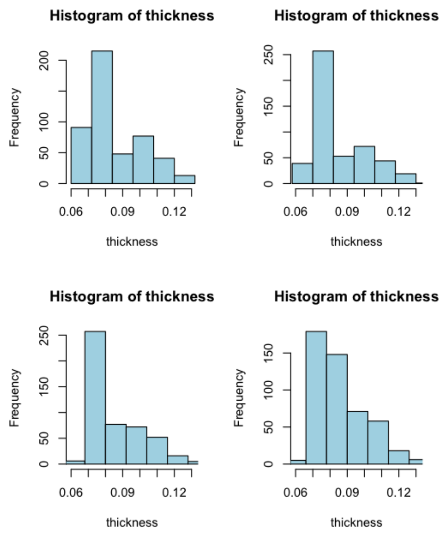
- $n = 485$ observed values of thickness for Mexico stamps
- over > 70 years
- During 1980s
- Stamp papers produced in several factories?
- No records. Can we guess by looking at the data?

Data Exploration - 1D Hidalgo Stamp

Histograms with different bin widths



Histograms with different bin locations (fixed $h = 0.012$)



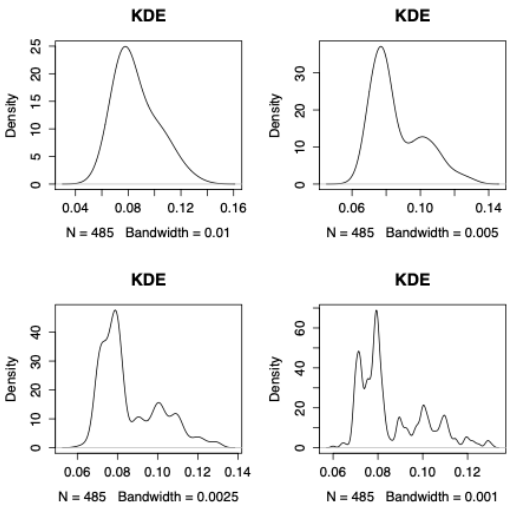
Data Exploration - Kernel density estimate

- Histograms are dependent on binwidth and bin location.
- Smaller binwidth preferred for exploration.
- Different bin locations can obscure important underlying patterns; A solution is to average out the effect on different bin location "Averaged histogram"
- For $X_1, ..., X_n \sim i.i.d.f(x)$ (continuous pdf), a kernel density estimator of f is obtained as

$$\hat{f}_n(x) = \frac{1}{n}\sum_{i=1}^{n}\frac{1}{h}K(\frac{x - X_i}{h})$$

where the kernel $K(.)$ is a function satisfying $\int K(x)dx = 1$.

Kernel density estimate (KDE) for Hidalgo Stamp



Data Exploration - Kernel density estimate

- Small bandwidth leads undersmooth; Large bandwidth leads oversmooth
- Is it unimodal? Bi-modal? Or several modes? Which modes are really there?
- Choice of bandwidth
  - Important in practice.
  - Controversial issue.
  - Many recommendations (Silverman's rule of thumb, Sheather-Jones Plug-In.)
  - The consensus is that there is never a consensus.

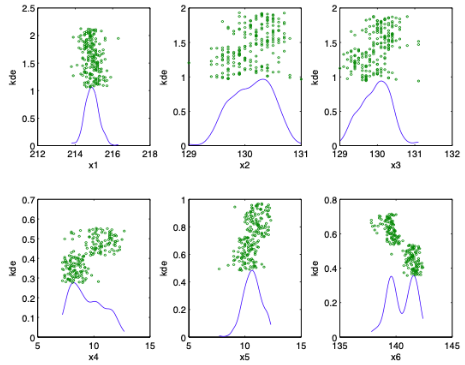
Example: Swiss bank notes

- $n = 200$ Swiss bank notes (See Fig. 1.1, Hardle and Simar)
- Each note (obs.) has $p = 6$ measurements (variables).
- Additional information: first half are genuine; the other half are counterfeit.
- Visualization of 6−dim'l data?
- Can use 6 KDEs overlaided with jitterplot for each measurements (variables)

Swiss bank notes - Marginal KDEs

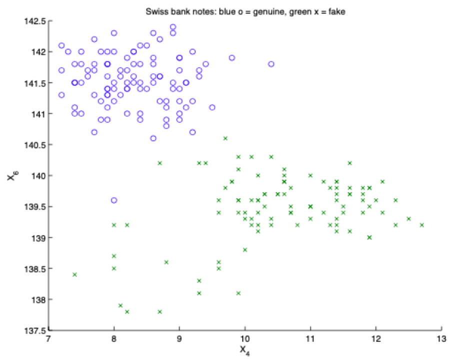
Marginal KDEs overlaided with jitterplot for each of 6 variables.

- Informative, realistic when p is small
- No information about association between variables.



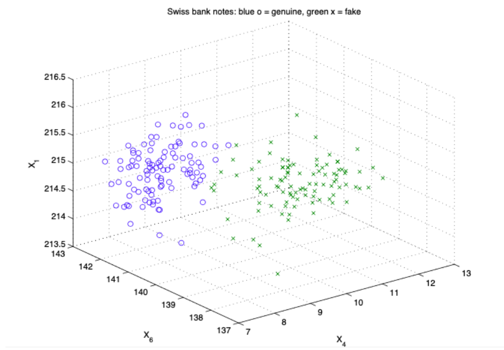
Swiss bank notes - Scatterplot

- Pairs of variables are best visualized by scatterplot, e.g. X_4 vs X_6 below.
- Understood as point clouds (representing the empirical distributions)
- C_6^2 many pairs to choose from.



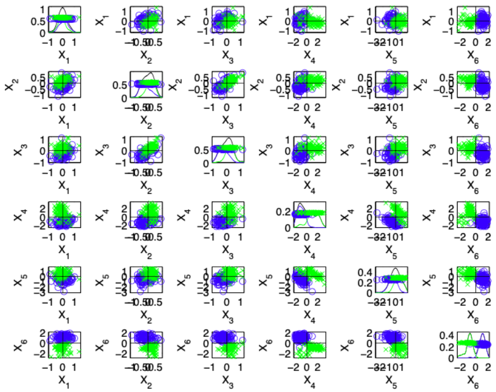
Swiss bank notes - Scatterplot

- Scatters of three variables can also be informative, if software allows to rotate the axes.
- Otherwise, the 3D scatterplot is a scatterplot of linear combinations of the three variables.



Swiss bank notes - Scatterplot matrix

- A traditional, yet powerful, tool is to construct a matrix of scatterplots. - Too busy with p = 6.



Distance Measures

- Many multivariate methods are based on Distance
- Distance can be measured from a p dimensional point $P = (x_1, \ldots, x_p)$ to:
  - The Origin $O = (0, \ldots, 0)$
  - Another (Fixed) Point $Q = (y_1, \ldots, y_p)$
- Distance measure can be based on:
  - Straight-Line (Euclidean) Distance - Equal weighting among the p variables that make up the point
  - Statistical Distance - Each variable scaled (divided) by its standard deviation
- Points that are equal distance from the origin make up:
  - Circles for Straight Line Distance
  - Ellipses for Statistical Distance

Straight-Line (Euclidean) Distance

- $p = 2$:

$$O = (0, 0) \quad P = (x_1, x_2) \quad Q = (y_1, y_2) \text{ (fixed point)}$$

- General p:

$$O = (0, \ldots, 0) \quad P = (x_1, \ldots, x_p) \quad Q = (y_1, \ldots, y_p) \text{ (fixed point)}$$

- Straight - Line (Euclidean):
  - $p = 2$:

$$d(O, P) = \sqrt{x_1^2 + x_2^2}$$

$$d(P, Q) = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2}$$

  - General p:

$$d(O, P) = \sqrt{x_1^2 + \cdots + x_p^2}$$

$$d(P, Q) = \sqrt{(x_1 - y_1)^2 + \cdots + (x_p - y_p)^2}$$

Statistical Distance

Statistical: $s_{kk} = s_k^2 = \frac{1}{n}\sum_{j=1}^{n}\left(x_{jk} - \overline{x}_k\right)^2 \quad ; \quad x_k^* = \frac{x_k}{\sqrt{s_{kk}}} \quad k = 1, \ldots, p$

- $p = 2$:

$$d(O, P) = \sqrt{\left(x_1^*\right)^2 + \left(x_2^*\right)^2} = \sqrt{\left(\frac{x_1}{\sqrt{s_{11}}}\right)^2 + \left(\frac{x_2}{\sqrt{s_{22}}}\right)^2} = \sqrt{\frac{x_1^2}{s_{11}} + \frac{x_2^2}{s_{22}}}$$

$$d(P, Q) = \sqrt{\frac{(x_1 - y_1)^2}{s_{11}} + \frac{(x_2 - y_2)^2}{s_{22}}}$$

- General p:

$$d(O, P) = \sqrt{\frac{x_1^2}{s_{11}} + \cdots + \frac{x_p^2}{s_{pp}}}$$

$$d(P, Q) = \sqrt{\frac{(x_1 - y_1)^2}{s_{11}} + \cdots + \frac{(x_p - y_p)^2}{s_{pp}}}$$

Statistical Distance

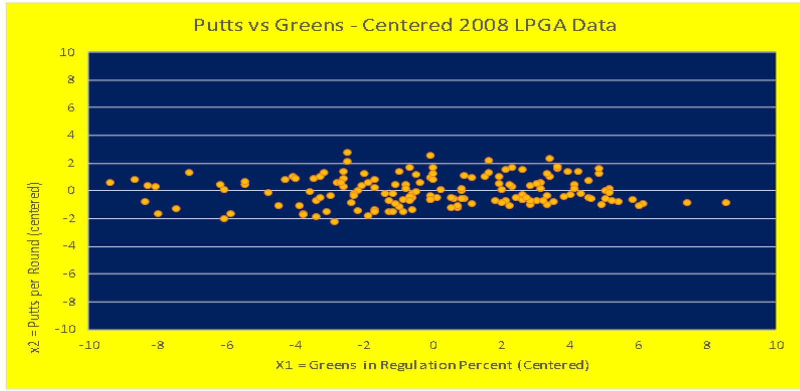
Set of Points with constant statistical distance from the Origin:

- $p = 2$:

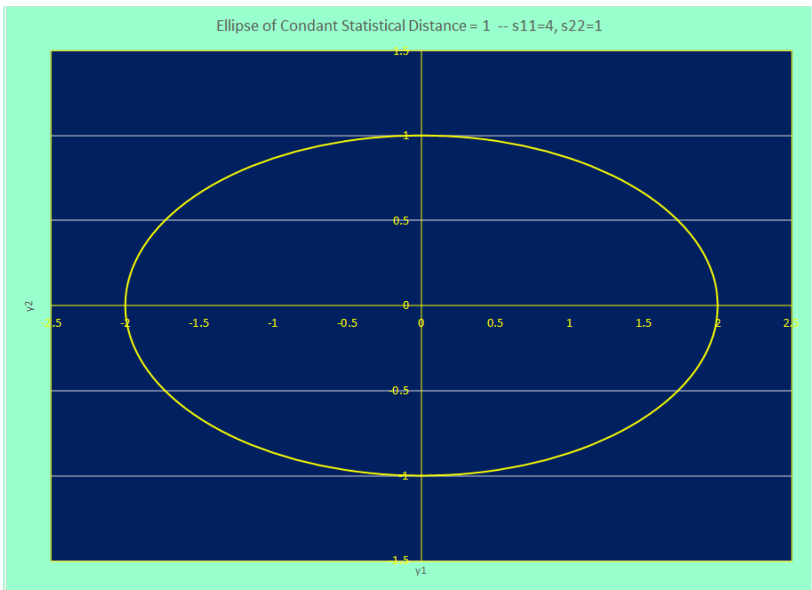
$$\frac{x_1^2}{s_{11}} + \frac{x_2^2}{s_{22}} = c^2$$

- General p:

$$\frac{x_1^2}{s_{11}} + \cdots + \frac{x_p^2}{s_{pp}} = c^2$$



Greens in Regulation:
Mean = 63.06 ; s11 = 12.95
Putts per Round:
Mean = 29.19 ; s22 = 1.07



Distance with Correlated Variables

With Correlated Variables, axes can be rotated by angle θ to better describe the variation

- New axes: $\tilde{x}_1, \tilde{x}_2$

$$\tilde{x}_1 = x_1 \cos(\theta) + x_2 \sin(\theta)$$

$$\tilde{x}_2 = -x_1 \sin(\theta) + x_2 \cos(\theta)$$

$p = 2$:

$$O = (0,0) \quad P = (x_1, x_2) \quad Q = (y_1, y_2) \text{ (fixed point)}$$

Distance with Correlated Variables

- Statistical: $\tilde{s}_{kk} = \frac{1}{n}\sum_{j=1}^{n}\left(\tilde{x}_{jk} - \overline{\tilde{x}}_k\right)^2 \qquad k = 1, ..., p$

$$p = 2: \quad d\left(O, P\right) = \sqrt{\frac{\tilde{x}_1^2}{s_{11}} + \frac{\tilde{x}_2^2}{\tilde{s}_{22}}} = \sqrt{a_{11}x_1^2 + 2a_{12}x_1x_2 + a_{22}x_2^2}$$

$$d\left(P, Q\right) = \sqrt{a_{11}(x_1 - y_1)^2 + 2a_{12}\left(x_1 - y_1\right)\left(x_2 - y_2\right) + a_{22}(x_2 - y_2)^2}$$

with

$$a_{11} = \frac{\cos^2(\theta)}{\cos^2(\theta)\, s_{11} + 2\sin(\theta)\cos(\theta)\, s_{12} + \sin^2(\theta)\, s_{22}} + \frac{\sin^2(\theta)}{\sin^2(\theta)\, s_{11} - 2\sin(\theta)\cos(\theta)\, s_{12} + \cos^2(\theta)\, s_{22}}$$

$$a_{12} = \frac{\cos(\theta)\sin(\theta)}{\cos^2(\theta)\, s_{11} + 2\sin(\theta)\cos(\theta)\, s_{12} + \sin^2(\theta)\, s_{22}} - \frac{\cos(\theta)\sin(\theta)}{\sin^2(\theta)\, s_{11} - 2\sin(\theta)\cos(\theta)\, s_{12} + \cos^2(\theta)\, s_{22}}$$

$$a_{22} = \frac{\sin^2(\theta)}{\cos^2(\theta)\, s_{11} + 2\sin(\theta)\cos(\theta)\, s_{12} + \sin^2(\theta)\, s_{22}} + \frac{\cos^2(\theta)}{\sin^2(\theta)\, s_{11} - 2\sin(\theta)\cos(\theta)\, s_{12} + \cos^2(\theta)\, s_{22}}$$

- Set of Points with constant statistical distance from Q:

$$p = 2: a_{11}(x_1 - y_1)^2 + 2a_{12}\left(x_1 - y_1\right)\left(x_2 - y_2\right) + a_{22}(x_2 - y_2)^2 = c^2$$

Data Ellipses Containing 80% of Measurements